

Part VIII — Electricity and Magnetism · Chapter 46

Electromagnetic Waves

How electric and magnetic fields sustain each other across empty space — and why light, radio, X-rays, and gamma rays are the same phenomenon

Sunlight reaches us across 150 million kilometres of vacuum. A radio signal from a spacecraft at the edge of the solar system takes hours to arrive, yet arrives as if no medium were needed for the journey. A hospital X-ray machine and a kitchen microwave oven both emit forms of light — invisible to the eye, but physically identical in mechanism to the golden glow of a sunset. The unification of these phenomena is one of the greatest achievements of classical physics: James Clerk Maxwell's prediction, in 1865, that changing electric and magnetic fields must propagate through empty space as a self-sustaining wave, travelling at exactly the speed of light. When Heinrich Hertz experimentally produced and detected such waves in 1888, he confirmed not only Maxwell's theory but also that light is electromagnetic. This chapter shows how electricity and magnetism, when they change, generate each other, and how that mutual generation creates waves that fill the universe and carry energy, information, and life-enabling radiation across all of space.

46.1 From Electricity and Magnetism to Electromagnetic Waves

Faraday's Law (Chapter 45) established that a changing magnetic field produces an electric field. Maxwell, in his mathematical synthesis of electromagnetism, recognised the necessity of a symmetric counterpart: a changing electric field must also produce a magnetic field. This addition — called the displacement current — completed the symmetry of the field equations and had an unexpected consequence.

If a changing magnetic field creates an electric field, and a changing electric field creates a magnetic field, then a self-sustaining oscillation becomes possible: the two fields continually recreate each other as they propagate through space. Neither field can exist alone in this oscillating state; each sustains the other. The result is an electromagnetic wave.

Historical Note: Maxwell completed his four equations of electromagnetism by 1865 and calculated the speed at which electromagnetic waves must travel through a vacuum. The speed depended on two constants of electrostatics and magnetostatics: the permittivity ϵ_0 and permeability μ_0 of free space. The result was $c = 1/\sqrt{\epsilon_0\mu_0} = 2.998 \times 10^8 \text{ m/s}$ — exactly the known speed of light. Maxwell immediately concluded that light is an electromagnetic wave. This was a purely theoretical prediction, made twenty-three years before Hertz produced radio waves in the laboratory. It remains one of the most powerful examples of mathematics revealing physical reality before experiment could confirm it.

Hertz's 1888 experiments produced oscillating electric sparks that generated radio waves, which he detected at a distance using a loop of wire. He showed that these waves reflected, refracted, and had a finite speed — exactly as Maxwell predicted. Hertz had no interest in practical applications; he said, "It's of no use whatsoever." Within a decade, Marconi had transmitted Morse code across the Atlantic. Within a century, electromagnetic waves of every kind had become the nervous system of global civilisation.

46.2 Changing Electric Fields and Changing Magnetic Fields

The key extension Maxwell made to Faraday's Law is the displacement current. In Faraday's Law: a changing magnetic field ($\Delta B/\Delta t$) generates a circulating electric field. Maxwell added the symmetric law: a changing electric field ($\Delta E/\Delta t$) generates a circulating magnetic field.

These two laws work together as a feedback system. Suppose we disturb the electric field at some location — for example, by oscillating a charge back and forth on an antenna. The changing E field generates a changing B field at nearby points in space. That changing B field generates a changing E field slightly further out. That new E field generates a new B field even further out. And so on, outward in all directions at the speed of light.

Key Point: The displacement current is not a current of charges. It is Maxwell's name for the effect of a changing electric field: just as a real current (moving charges) generates a magnetic field, a changing electric field generates the same kind of magnetic field as if a current were flowing. This insight closed the mathematical inconsistency in Ampère's Law and simultaneously predicted electromagnetic waves.

This propagation does not require any medium — no air, no water, no ether (a medium that was once hypothesised but shown not to exist; see Chapter 47). The fields carry themselves through empty space, each producing the other, advancing at $c = 2.998 \times 10^8 \text{ m/s}$.

46.3 Self-Sustaining Field Oscillations

In a transverse electromagnetic wave, the electric field E oscillates in one direction, the magnetic field B oscillates in a perpendicular direction, and the wave travels in the third perpendicular direction. All three are mutually perpendicular: if the wave travels in the x direction, E might oscillate in the y direction and B in the z direction.

The amplitude of E and B are related by:

$$E / B = c$$

In a vacuum, the ratio of the electric field amplitude to the magnetic field amplitude in an electromagnetic wave equals the speed of light $c \approx 3.00 \times 10^8$ m/s. This ratio holds at every point in the wave at every instant.

Both fields oscillate in phase: they peak together and pass through zero together. The wave is self-sustaining because when E is changing most rapidly (at the zero-crossing), it is generating B most effectively, and vice versa. The wave carries energy in both the E and B fields equally.

Electromagnetic waves are transverse waves: the oscillations of E and B are perpendicular to the direction of propagation. They are not longitudinal (compression) waves like sound. This is why electromagnetic waves can be polarised — E can be made to oscillate in a fixed plane (linear polarisation), as in polarised sunglasses and LCD screens.

46.4 Electromagnetic Waves Do Not Require a Medium

Sound waves require a mechanical medium (solid, liquid, or gas) to travel. Sound cannot propagate in a vacuum. Electromagnetic waves require no medium at all. They travel through empty space just as well as through air — better, in fact, since some materials absorb or slow them.

This was a revolutionary and initially controversial idea. Nineteenth-century physicists assumed that all waves required a medium, and they postulated a medium called the “luminiferous ether” to carry light waves. The Michelson–Morley experiment (1887) searched for evidence of the ether and found none. The failure of ether theory is discussed in detail in Chapter 47; its resolution required Einstein’s special relativity.

The modern understanding is that the electromagnetic field itself is the physical entity. The oscillating fields are real: they carry momentum and energy. They do not need a medium to exist in; they are not disturbances of anything. They are disturbances of the electromagnetic field, and the electromagnetic field is a fundamental feature of space.

The practical consequence: electromagnetic radiation can cross the vacuum of space

between galaxies, billions of light-years, without requiring anything to be present along the way. Radio waves from the Big Bang itself (the cosmic microwave background) have been propagating through expanding space for 13.8 billion years and are detectable by sensitive telescopes today.

46.5 Speed of Light and Wave Speed

In a vacuum, all electromagnetic waves travel at the same speed, regardless of frequency:

$$c = 2.998 \times 10^8 \text{ m/s} \approx 3.00 \times 10^8 \text{ m/s}$$

The speed of light in vacuum, c , is a universal constant. It is defined exactly as 299 792 458 m/s. The kilogram, metre, and second are now defined such that c has this exact value.

In a material medium (glass, water, air), electromagnetic waves travel slower than c . The ratio of c to the wave speed in the medium is the refractive index n : $v = c/n$. For air, $n \approx 1.0003$, so light slows only slightly. For glass, $n \approx 1.5$, so light travels at about 2/3 of its vacuum speed in glass. For water, $n \approx 1.33$.

The wave relationship for electromagnetic waves is the same as for all waves:

$$c = f \times \lambda$$

Wave speed equation. c = speed of light ($3.00 \times 10^8 \text{ m/s}$ in vacuum); f = frequency (Hz); λ = wavelength (m). Since c is constant in vacuum, high-frequency waves have short wavelengths, and low-frequency waves have long wavelengths.

Example — Wavelength of visible light Green light has a frequency of approximately $5.5 \times 10^{14} \text{ Hz}$. $\lambda = c/f = 3.00 \times 10^8 / 5.5 \times 10^{14} = 5.45 \times 10^{-7} \text{ m} = 545 \text{ nm}$. This is within the visible range (400–700 nm).

Example — Frequency of FM radio An FM radio station broadcasts at wavelength 3.0 m. $f = c/\lambda = 3.00 \times 10^8 / 3.0 = 1.0 \times 10^8 \text{ Hz} = 100 \text{ MHz}$. This is in the FM radio band (88–108 MHz).

46.6 Wavelength, Frequency, and Energy

The energy carried by an electromagnetic wave is related to both its intensity (energy per unit area per unit time) and the frequency of the radiation. In the classical (wave) picture, the energy is proportional to the square of the amplitude of E (or B). In the quantum picture (relevant to Chapters 48 onward), electromagnetic radiation arrives in discrete packets called photons, each with energy:

$$E = hf$$

Photon energy equation (introduced here for completeness; developed fully in Chapter 48). E = energy of one photon (joules); h = Planck's constant = 6.626×10^{-34} J·s; f = frequency (Hz). Higher frequency = higher energy per photon.

This equation has profound consequences for biological and technological interactions with radiation. High-frequency radiation (UV, X-ray, gamma) carries high energy per photon and can break chemical bonds, ionise atoms, and damage DNA. Low-frequency radiation (radio waves, microwaves) carries low energy per photon and generally cannot ionise atoms or break bonds.

For this chapter, the wave description suffices: high-frequency electromagnetic radiation has short wavelengths and high photon energies; low-frequency radiation has long wavelengths and low photon energies. The ordering from low frequency to high frequency defines the electromagnetic spectrum.

46.7 The Electromagnetic Spectrum

The electromagnetic spectrum is the complete range of all electromagnetic radiation, ordered by frequency (or equivalently by wavelength or photon energy). There are no gaps and no hard boundaries between named regions — the names are human conventions for ranges that have different practical interactions with matter.

Region	Wavelength Range	Frequency Range	Photon Energy	Example Sources
Radio waves	> 1 mm (up to km)	< 300 GHz	< 10^{-12} eV	Antennas, lightning, pulsars
Microwaves	1 mm – 30 cm	1–30 GHz (approx)	meV range	Microwave ovens, radar, CMB
Infrared (IR)	700 nm – 1 mm	300 GHz – 430 THz	1–20 meV	Warm objects, IR heaters, remote controls
Visible light	400–700 nm	430–750 THz	1.8–3.1 eV	Stars, LEDs, fire
Ultraviolet (UV)	10–400 nm	750 THz – 30 PHz	3–124 eV	Sun, arc lamps, sterilisers
X-rays	0.01–10 nm	30 PHz – 30 EHz	124 eV – 124 keV	X-ray tubes, synchrotrons
Gamma rays (γ)	< 0.01 nm	> 30 EHz	> 100 keV	Radioactive nuclei, pulsars, GRBs

Table 46.1 *The electromagnetic spectrum. Boundaries are approximate and conventional. All regions are the same physical phenomenon at different frequencies.*

Within the visible range, colour corresponds to wavelength: violet (≈ 400 – 450 nm), blue (450 – 490 nm), green (490 – 570 nm), yellow (570 – 590 nm), orange (590 – 630 nm), red (630 – 700 nm). The human eye is sensitive to this narrow range because the Sun emits most intensely in the visible band and our visual system evolved to exploit it.

46.8 Radio Waves and Communication

Radio waves span the largest range of the electromagnetic spectrum, from wavelengths of millimetres up to kilometres. They interact weakly with most matter, penetrating walls, clouds, and the ionosphere under many conditions. They are generated by oscillating electric currents in antennas and detected by the currents they induce in receiving antennas.

Radio communication works by impressing information onto a carrier wave through modulation. Amplitude modulation (AM) varies the amplitude of the carrier wave in proportion to the audio signal. Frequency modulation (FM) varies the frequency slightly. FM is less susceptible to static (which affects amplitude) and gives better audio fidelity, which is why FM is preferred for music broadcasting.

Different frequency bands of the radio spectrum are allocated for different purposes by international agreement. AM radio uses 540–1600 kHz (wavelengths 190–560 m). FM radio uses 88–108 MHz (wavelengths 2.8–3.4 m). Television uses VHF and UHF bands. Mobile phones use 700 MHz – 3 GHz. WiFi uses 2.4 GHz and 5 GHz. GPS uses 1.2–1.6 GHz.

Long-wave radio (30 kHz – 3 MHz) can diffract around the Earth's curvature, enabling long-distance communication without line-of-sight. Short-wave radio (3–30 MHz) can reflect from the ionosphere, also enabling long-distance propagation. Higher-frequency radio and microwave signals travel in straight lines and require relay stations, satellites, or fibre-optic cables for global communication.

46.9 Microwaves and Radar

Microwaves occupy wavelengths from about 1 mm to 30 cm, roughly 1–300 GHz. They are generated by specialised vacuum tubes (magnetrons, klystrons) and solid-state sources.

Radar (Radio Detection And Ranging) uses microwave pulses to locate objects. A pulse is transmitted; when it reflects off an object, the return signal is timed. Since the wave travels at c , the distance to the object is $d = ct/2$, where t is the round-trip travel time. Radar is used for aircraft tracking, weather monitoring, ship navigation, and speed enforcement cameras. The Doppler shift of the reflected pulse (a change in frequency due to the relative motion of the target) allows speed measurement.

Microwave ovens exploit a different property: water molecules in food absorb microwave radiation (at 2.45 GHz) and convert it to thermal energy by rotating in response to the oscillating electric field. The microwave radiation does not heat the

container (if non-metallic) because containers lack the rotating dipoles that absorb microwave energy efficiently. The energy absorbed by the food heats it from within, rather than from the surface inward as in conventional ovens.

The cosmic microwave background (CMB) is microwave radiation arriving from all directions in space, the redshifted remnant of the intense radiation field of the early universe about 380 000 years after the Big Bang. Its temperature is 2.725 K and its spectrum is the most perfect blackbody spectrum ever measured. The CMB is strong evidence for the Big Bang model and provides a wealth of information about the early universe.

46.10 Infrared, Visible Light, and Ultraviolet

Infrared radiation (IR) has wavelengths from about 700 nm to 1 mm. All objects above absolute zero emit infrared radiation; hotter objects emit more intensely and at shorter wavelengths (Wien's Law, Chapter 47). Human body temperature ($37^{\circ}\text{C} = 310\text{ K}$) corresponds to peak emission around 9400 nm, well into the infrared. Thermal imaging cameras detect IR and can see people in darkness, find heat leaks in buildings, and guide missiles toward warm engines.

TV remote controls transmit simple commands using pulsed IR at about 940 nm. Near-IR radiation (700–2500 nm) is used in fibre-optic telecommunications: glass fibres have very low absorption at these wavelengths, allowing signals to travel many kilometres with little loss.

Visible light occupies 400–700 nm. This is the narrow band to which the human eye responds. The Sun emits across a broad spectrum but peaks in the visible (a green-yellow colour at about 550 nm). The atmosphere absorbs many UV and IR wavelengths, so the visible band reaching the Earth's surface is particularly rich.

Ultraviolet (UV) radiation has wavelengths 10–400 nm. The Sun emits substantial UV; most UV-B (280–315 nm) and all UV-C (100–280 nm) are absorbed by atmospheric oxygen and ozone before reaching the surface, protecting life. UV-A (315–400 nm) does reach the surface and causes tanning and contributes to skin ageing and cancer. UV-C is germicidal: it damages DNA and is used to sterilise water, air, and surfaces. High-energy UV can ionise atoms and break chemical bonds, making prolonged exposure biologically damaging.

46.11 X-Rays and Gamma Rays

X-rays have wavelengths from about 0.01 to 10 nm and photon energies of 100 eV to 100 keV. They are generated by accelerating electrons through high voltages and directing them at a metal target; the sudden deceleration produces a broad spectrum of X-rays (bremsstrahlung), along with characteristic X-rays at specific energies that depend on the target material.

X-rays penetrate soft tissue but are partially absorbed by denser materials such as bone and metal. This differential absorption makes them invaluable for medical imaging: an X-ray image shows the shadows of dense structures (bones, tumours, foreign objects) against the less-absorbing background of soft tissue. CT (computed tomography) scans use many X-ray images from different angles and computer reconstruction to produce three-dimensional images.

X-rays also have technological uses outside medicine. X-ray crystallography uses the diffraction of X-rays by crystal lattices to determine atomic-scale structures. The structure of DNA (the double helix) was solved in 1953 using X-ray crystallography data. Industrial X-ray inspection checks welds and castings for internal defects without disassembly. Airport security scanners use X-rays to image the contents of luggage.

Gamma rays (γ -rays) have wavelengths below about 0.01 nm and photon energies above about 100 keV. They are produced by nuclear transitions (radioactive decay, Chapter 51) and by extreme astrophysical events (gamma-ray bursts, pulsars, neutron star mergers). Gamma rays are the most penetrating form of electromagnetic radiation and require thick shielding of lead or concrete. They are used in medicine for radiotherapy (targeting tumours with high-energy radiation) and in diagnostic imaging (PET scans use gamma rays from positron annihilation).

Key Point: X-rays and gamma rays are distinguished by their origin, not strictly by their photon energy. X-rays originate from electron transitions (either in atoms or from deceleration), while gamma rays originate from nuclear transitions. Some X-ray photons have higher energies than some gamma-ray photons, so the distinction is source-based, not energy-based.

46.12 Energy Transfer by Electromagnetic Radiation

Electromagnetic waves carry energy through space. The intensity I of electromagnetic radiation is the power per unit area delivered to a surface perpendicular to the direction of propagation:

$$I = P / A$$

Intensity (W/m^2) = power P (W) $\div$ area A (m^2). Also equal to $\epsilon_0 c E^2$ for the wave amplitude form. Intensity decreases with the square of the distance from a point source: $I \propto 1/r^2$ (inverse square law).

For a point source (emitting equally in all directions), the radiation spreads over a sphere of area $4\pi r^2$. Intensity therefore falls as $1/r^2$ with distance r from the source. This is the inverse square law of radiation, and it applies to starlight, sound from a speaker, and any other wave radiating from a compact source.

$$I_2/I_1 = (r_1/r_2)^2$$

Inverse square law for intensity. If distance doubles ($r_2 = 2r_1$), intensity drops to one quarter. If distance halves, intensity quadruples.

Example — Solar irradiance The Sun delivers about 1361 W/m^2 at Earth's distance ($1 \text{ AU} \approx 1.50 \times 10^{11} \text{ m}$). Find the total power output of the Sun. $P = I \times 4\pi r^2 = 1361 \times 4\pi \times (1.50 \times 10^{11})^2 = 1361 \times 2.83 \times 10^{23} \approx 3.85 \times 10^{26} \text{ W}$. This is the Sun's luminosity — about 3.85×10^{26} watts emitted continuously as electromagnetic radiation.

Electromagnetic radiation also carries momentum. When absorbed, it exerts a radiation pressure on a surface. This pressure is tiny under normal conditions but is measurable and has engineering significance: solar sails proposed for spacecraft propulsion would use radiation pressure from sunlight. Radiation pressure is also significant in very high-energy astrophysical environments.

46.13 Safety, Shielding, and Biological Effects

The biological effects of electromagnetic radiation depend strongly on frequency (and thus photon energy). The division is broadly between non-ionising radiation (insufficient energy per photon to ionise atoms) and ionising radiation (sufficient energy to remove electrons from atoms, breaking chemical bonds and damaging DNA).

Type	Ionising?	Biological Risk	Shielding / Mitigation
Radio, microwave	No	Thermal heating at very high intensities	Limit exposure time; distance
Infrared	No	Burns from intense IR (IR lasers, furnaces)	Reflective clothing, IR goggles
Visible light	No (mostly)	Retinal damage from lasers	Laser safety goggles
Ultraviolet (UV-A/B)	Borderline	Sunburn, DNA damage, skin cancer, cataracts	Sunscreen, UV-blocking glass, clothing
UV-C, X-rays	Yes	DNA damage, cell death, cancer risk	Distance, lead shielding, time limits
Gamma rays	Yes (highly)	Severe ionisation, radiation sickness, death	Thick lead or concrete; remote handling

Table 46.2 *Biological effects and shielding strategies for electromagnetic radiation.*

The non-ionising/ionising boundary falls roughly at the UV region: UV-B and UV-C photons (energies above 3.9 eV) carry enough energy to break the hydrogen bonds and covalent bonds in DNA. X-rays and gamma rays are strongly ionising and are regulated as sources of radiation exposure.

Microwave radiation at household oven intensities is non-ionising: individual photons carry insufficient energy to break chemical bonds. The only biological effect of microwave radiation is heating (dielectric heating of polar molecules, especially water). At the intensities used by mobile phones and WiFi routers (milliwatts), even the heating effect is negligible and no mechanism for DNA damage exists. Decades of epidemiological studies have not established a causal link between exposure to low-intensity radio/microwave radiation and any adverse health outcome. The scientific consensus is that such radiation at permitted exposure levels is safe.

46.14 Common Mistakes in Electromagnetic Wave Analysis

- Confusing wavelength and frequency. They are inversely related: $\lambda = c/f$. Higher frequency = shorter wavelength. Gamma rays have the highest frequency and shortest wavelength; radio waves have the lowest frequency and longest wavelength. Drawing the spectrum in the wrong order is a common error.
- Thinking light requires a medium. Electromagnetic waves do not require a medium; they propagate through vacuum. Sound waves do require a medium. Students who know “waves require a medium” must remember that this rule applies to mechanical waves (sound, water waves, seismic waves) but not to electromagnetic waves.
- Applying $v = f\lambda$ with the wrong speed. In vacuum, use $c = 3.00 \times 10^8$ m/s. In a medium, use $v = c/n < c$. If you assume vacuum speed in glass and calculate a wavelength, you will get the vacuum wavelength, not the in-glass wavelength. The frequency does not change when light enters a medium; the wavelength does.
- Assuming all electromagnetic waves have the same energy. The wave amplitude determines intensity; the frequency determines photon energy. A very bright infrared source is less dangerous to DNA than faint UV, because visible/IR photons carry insufficient energy per photon to break bonds, regardless of how many arrive.
- Confusing ionising and non-ionising radiation. Radio waves and microwaves are non-ionising. X-rays and gamma rays are ionising. UV sits at the boundary. Many students incorrectly categorise all radiation as equally hazardous or equally safe.
- Applying the inverse square law to beams. The inverse square law ($I \propto 1/r^2$) applies to point sources emitting in all directions. A laser beam is highly collimated and does not spread isotropically; its intensity stays approximately constant over its beam width. Radio towers, light bulbs, and stars are approximately point sources; lasers and directional antennas are not.
- Forgetting that c is constant in vacuum for all frequencies. In vacuum, radio waves, visible light, X-rays, and gamma rays all travel at exactly $c = 2.998 \times 10^8$ m/s. The speed does not depend on frequency. (In a medium, different frequencies may travel at slightly different speeds — this is dispersion, which causes a prism to separate colours — but in vacuum all frequencies travel identically.)

Common Mistake: When calculating frequency from wavelength (or vice versa), always confirm which medium the wave is in. In air ($n \approx 1.0003$), using c is fine. In glass, water, or other materials, use $v = c/n$. Confusing the two leads to frequency errors, which then cascade into energy errors.

Chapter Summary

Electromagnetic waves are self-sustaining oscillations of electric and magnetic fields that propagate through vacuum at $c \approx 3.00 \times 10^8$ m/s. They require no medium and carry energy and momentum.

- Maxwell predicted EM waves by showing that changing E fields generate B fields (extending Ampère's Law) and changing B fields generate E fields (Faraday's Law). Hertz confirmed them experimentally in 1888.
 - In an EM wave: E and B are perpendicular to each other and to the direction of propagation; $E/B = c$; both fields oscillate in phase.
 - Wave equation: $c = f\lambda$. In vacuum, c is constant for all frequencies.
 - Photon energy: $E = hf$. Higher frequency = higher energy per photon.
 - The electromagnetic spectrum (low to high frequency): radio, microwave, infrared, visible, UV, X-ray, gamma ray. All are the same phenomenon.
 - Intensity follows the inverse square law from point sources: $I \propto 1/r^2$.
 - Biological effects depend on frequency: non-ionising (radio through near-UV) vs. ionising (far-UV through gamma). Ionising radiation can break chemical bonds and damage DNA.
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Key Terms

electromagnetic wave: A self-sustaining oscillation of mutually perpendicular E and B fields propagating through space at speed c ; requires no medium.

speed of light (c): The speed of all electromagnetic waves in vacuum: $c = 2.998 \times 10^8$ m/s, defined exactly as 299 792 458 m/s.

electromagnetic spectrum: The complete range of electromagnetic radiation ordered by frequency; includes radio, microwave, infrared, visible, UV, X-ray, and gamma rays.

wavelength (λ): The distance between successive crests of a wave; related to frequency by $\lambda = c/f$; SI unit metre (m).

frequency (f): The number of wave cycles per second; SI unit hertz ($\text{Hz} = \text{s}^{-1}$); related to wavelength by $f = c/\lambda$.

photon energy ($E = hf$): The energy carried by one quantum (photon) of electromagnetic radiation; $E = hf$; higher frequency = higher photon energy.

Planck's constant (h): $h = 6.626 \times 10^{-34}$ J·s; the constant of proportionality between photon frequency and photon energy.

intensity (I): Power per unit area delivered by a wave: $I = P/A$ (W/m^2); falls as $1/r^2$ from a point source (inverse square law).

inverse square law: The intensity of radiation from a point source is inversely proportional to the square of the distance: $I \propto 1/r^2$.

ionising radiation: Electromagnetic radiation with photon energies sufficient to ionise atoms or break chemical bonds; includes UV-C, X-rays, and gamma rays.

non-ionising radiation: Electromagnetic radiation with photon energies insufficient to ionise atoms; includes radio waves, microwaves, infrared, visible, and UV-A/B.

displacement current: Maxwell's term for the magnetic-field-generating effect of a changing electric field; the symmetric counterpart to Faraday's Law.

refractive index (n): $n = c/v$; the ratio of the speed of light in vacuum to its speed in a medium; $n \geq 1$ for all real materials.

Concept Review Questions

Answer in complete sentences.

1. Maxwell extended Ampère's Law by adding the displacement current. Explain in your own words what this addition means physically and why it was necessary to make the field equations mathematically consistent.
 2. Explain why electromagnetic waves do not require a medium. How does this differ from sound waves? What does the failure of the luminiferous ether theory suggest?
 3. In an electromagnetic wave, the E and B fields are always perpendicular to each other and to the direction of travel. Why can't both fields oscillate in the same direction as the wave travels?
 4. Radio waves and gamma rays are both electromagnetic radiation. List three properties they share and two properties that differ.
 5. Why does high-frequency UV light damage skin while the much more intense light from a warm fire (mostly infrared) does not damage DNA? Use the concept of photon energy.
 6. Explain why mobile phone signals can penetrate building walls while visible light cannot. What property of radiation determines its ability to penetrate solid materials?
 7. A star is 4 light-years away. If the star suddenly brightens, how long before we notice? What does this tell us about the nature of electromagnetic waves?
 8. Explain why microwave ovens heat food containing water more efficiently than food with no water content. What property of the water molecule is responsible?
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Spectrum Identification Exercises

For each description, identify the region of the electromagnetic spectrum.

9. Wavelength 550 nm. Name the region and, if visible, the approximate colour.
 10. Frequency 98 MHz.
 11. Photon energy 150 keV.
 12. Wavelength 10 nm.
 13. Wavelength 2.5 cm. Used to detect aircraft.
 14. Emitted by the human body at rest; used in thermal cameras.
 15. Produced by nuclear decay; most penetrating EM radiation.
 16. Wavelength 350 nm; causes tanning and sunburn.
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Wave Speed Problems

17. A radio station broadcasts at 720 kHz (AM). Find the wavelength in air.
 18. The wavelength of a laser beam in vacuum is 632 nm. (a) Find its frequency. (b) Find its wavelength in glass ($n = 1.50$). (c) Does its frequency change in glass?
 19. A radar pulse is transmitted and the reflection is received 80 μs later. How far away is the object?
 20. A WiFi router operates at 5.0 GHz. Find the wavelength in air. Compare to the wavelength of AM radio at 1.0 MHz.
 21. The cosmic microwave background has a peak wavelength of about 1.9 mm. Find its frequency. What region of the spectrum is this?
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Electromagnetic Radiation Applications

22. A 100 W incandescent bulb emits uniformly in all directions. (a) Find the intensity 1.0 m from the bulb. (b) Find the intensity 3.0 m from the bulb. (c) At what distance does the intensity drop to 1.0 W/m^2 ?
23. The Sun has luminosity $3.85 \times 10^{26} \text{ W}$. (a) Find the solar irradiance at Earth ($1.50 \times 10^{11} \text{ m}$ from the Sun). (b) Find the solar irradiance at Mars ($2.28 \times 10^{11} \text{ m}$). (c) How does this affect temperature on Mars relative to Earth?
24. An X-ray photon has energy 80 keV. (a) Find its frequency. (b) Find its wavelength. (c) Is this more or less energetic than a photon of green light (550 nm)?
25. A UV sterilisation lamp emits 15 W at 254 nm. (a) Find the energy per photon. (b) Find the number of photons emitted per second. (c) Explain why UV-C at 254 nm kills bacteria but infrared at 1000 nm of the same power does not.

Communication Technology Analysis

26. Explain the difference between AM and FM radio modulation. Why does FM provide better sound quality in most conditions?
27. A geostationary satellite orbits at 35 786 km above Earth's equator. (a) Find the signal travel time (one way) from the satellite to a ground station. (b) Find the round-trip delay. (c) Why does this delay cause problems for interactive voice calls via geostationary satellite, but not for television broadcasting?
28. Optical fibre carries infrared light at 1550 nm in glass ($n = 1.47$). (a) Find the speed of light in the fibre. (b) Find the frequency of the 1550 nm light. (c) What is the travel time for a signal to travel 1000 km through this fibre?
29. Compare the penetration properties of the following for communication inside buildings: (a) 100 MHz (FM radio), (b) 2.4 GHz (WiFi), (c) 60 GHz (proposed high-speed wireless). Explain in terms of wavelength and interaction with materials. Which would you expect to penetrate walls most easily?

Chapter 46 Checkpoint

Before moving on to Chapter 47, confirm you can do each of the following.

I can...	See Section
Explain how changing E and B fields sustain each other and produce an EM wave	46.1–46.3
State that EM waves require no medium and travel at c in vacuum	46.4–46.5
Apply $c = f\lambda$ to calculate wavelength or frequency	46.5
State $E = hf$ and explain how photon energy varies across the spectrum	46.6
Name the regions of the EM spectrum in order of frequency and wavelength	46.7
Describe uses and properties of radio waves, microwaves, IR, and visible light	46.8–46.10
Describe the production, uses, and risks of X-rays and gamma rays	46.11
Apply the inverse square law $I \propto 1/r^2$ to intensity calculations	46.12
Distinguish ionising from non-ionising radiation and explain biological effects	46.13
Avoid common mistakes in spectrum identification and wave calculations	46.14

Continue to Chapter 47: Why Modern Physics Was Needed →